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G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech

TERM-I (Regular)

CAE I- WINTER 2025 (2024 Pattern)

Matrices and Differential Calculus (25UBSL1101)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Find Cofactor of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 2 & -1 \end{bmatrix}$	[5]	1	2
b)	Define Rank of Matrix. Find the Rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 1 & 2 \\ -1 & -2 & -3 & -1 & -2 \\ 0 & 2 & 2 & 3 & 2 \\ 2 & 4 & 6 & 2 & 4 \end{bmatrix}$	[5]	1	3
	OR			
Q 2. (a)	Find A^{-1} by using Adjoint method where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 3 & 2 & -1 \end{bmatrix}$	[5]	1	2
(b)	Check whether the system has a unique solution. If exists then find the solution of $x + y + z = 2$ $x + y + 2z = 2$ $x - y + z = 4$	[5]	1	3
Q 3. (a)	Reduced the given matrix to diagonal form ($P^{-1}AP = D$) $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & 3 \end{bmatrix}$	[5]	2	3
(b)	1) If 3 and 15 are the two eigen values of A $= \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, find $\lvert A \rvert$, without expanding the determinant. 2) Find the eigen values of A $= \begin{bmatrix} 4 & 6 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 8 \end{bmatrix}$	[5]	2	3
	OR			
Q. 4 (a)	Find The Eigen Value and Eigen vectors of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	[5]	2	3
(b)	State and Verify Cayley Hamilton Theorem For $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$ and hence find A^{-1} by using characteristic polynomials.	[5]	2	3